

A Hilbert-Subspace Spaceability Result for Non-Differentiable Pettis Primitives

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2026-07-01

Source Problem

Bongiorno–Darji–Di Piazza [1] prove that, for every infinite dimensional Banach space X , the set $\mathcal{ND}_X$ of strongly measurable Pettis integrable functions $f: [0, 1] \rightarrow X$ whose Pettis primitive is nowhere weakly differentiable is lineable. They ask whether $\mathcal{ND}_X$ is spaceable.

We use the standard Pettis norm

$$\|f\|_{\mathcal{P}} = \sup_{E \in \mathcal{L}} \left\| (P) \int_E f d\lambda \right\|,$$

with functions identified in the usual Pettis-norm sense. The source paper explicitly discusses this norm topology and notes that it is not complete in general.

Partial Result

Theorem 1. *Let X be a Banach space containing a subspace isomorphic to ℓ_2 . Then $\mathcal{ND}_X$ is spaceable in the Pettis norm topology.*

Idea of the Proof

The source paper’s ℓ_2 -case uses dyadic intervals, pairwise disjoint positive-measure sets $A(\sigma, i) \subset I_\sigma$, and an orthonormal family $e(\sigma, i)$. Its lineability argument chooses many branches whose coordinate choices are eventually almost disjoint.

For closedness, we instead put the n th generator permanently in the coordinate column $i = n$, beginning at dyadic level n . A factor $\sqrt{n+1}$ exactly compensates for the tail $\sum_{k \geq n} (k+1)^{-2}$. This makes the Pettis norm of the generated function equivalent to the ℓ_2 -norm of the coefficient sequence, hence the generated subspace is closed. The first nonzero coefficient still appears on every sufficiently fine dyadic scale, forcing unbounded primitive difference quotients at every point.

Proof

We first prove the result for $X = \ell_2$. Use the notation from the source paper’s Section 3. Thus $2^{<\infty}$ is the dyadic tree, I_σ is the dyadic interval associated to σ , and, for every pair (σ, i) with $0 \leq i \leq |\sigma|$, $A(\sigma, i) \subset I_\sigma$ are pairwise disjoint measurable sets of positive measure. Let $\{e(\sigma, i)\}$ be the corresponding orthonormal basis of ℓ_2 .

For $n \geq 0$, define

$$g_n = \sqrt{n+1} \sum_{k=n}^{\infty} \sum_{\sigma \in 2^k} \frac{1}{(k+1)2^{k/2}} \frac{1}{\lambda(A(\sigma, n))} e(\sigma, n) \chi_{A(\sigma, n)}.$$

For $a = (a_n) \in \ell_2$, set

$$G_a = \sum_{n=0}^{\infty} a_n g_n.$$

Equivalently,

$$G_a = \sum_{k=0}^{\infty} \sum_{\sigma \in 2^k} \sum_{n=0}^k a_n \sqrt{n+1} \frac{1}{(k+1)2^{k/2}} \frac{1}{\lambda(A(\sigma, n))} e(\sigma, n) \chi_{A(\sigma, n)}.$$

For a measurable set E , write

$$\theta_{\sigma, n}(E) = \frac{\lambda(E \cap A(\sigma, n))}{\lambda(A(\sigma, n))}, \quad 0 \leq \theta_{\sigma, n}(E) \leq 1.$$

Then

$$(P) \int_E G_a = \sum_{k=0}^{\infty} \sum_{\sigma \in 2^k} \sum_{n=0}^k a_n \sqrt{n+1} \frac{\theta_{\sigma, n}(E)}{(k+1)2^{k/2}} e(\sigma, n).$$

By orthogonality,

$$\left\| (P) \int_E G_a \right\|_2^2 \leq \sum_{k=0}^{\infty} \sum_{\sigma \in 2^k} \sum_{n=0}^k \frac{|a_n|^2 (n+1)}{(k+1)^2 2^k} = \sum_{n=0}^{\infty} |a_n|^2 (n+1) \sum_{k=n}^{\infty} \frac{1}{(k+1)^2}.$$

For every $n \geq 0$,

$$1 \leq (n+1) \sum_{k=n}^{\infty} \frac{1}{(k+1)^2} \leq 2.$$

Hence

$$\|G_a\|_{\mathcal{P}} \leq \sqrt{2} \|a\|_2.$$

Taking $E = [0, 1]$ gives equality in the coefficient factors $\theta_{\sigma, n}(E) = 1$, so the same estimate also yields

$$\|G_a\|_{\mathcal{P}} \geq \left\| (P) \int_{[0, 1]} G_a \right\|_2 \geq \|a\|_2.$$

Thus $T: a \mapsto G_a$ is an isomorphic embedding of ℓ_2 into the Pettis-normed space $\mathcal{P}([0, 1], \ell_2)$. Since ℓ_2 is complete and the induced norm is equivalent to $\|\cdot\|_2$, $T(\ell_2)$ is closed in the Pettis norm.

It remains to check that every nonzero G_a belongs to $\mathcal{ND}_{\ell_2}$. Let $a \neq 0$, and let n_0 be the least index with $a_{n_0} \neq 0$. Let $F_a(t) = (P) \int_0^t G_a$. Fix $x \in [0, 1]$ and $M > 0$. For every sufficiently small $h > 0$, choose a dyadic interval $I_\tau \subset [x, x+h]$ of length 2^{-j} , with $j \geq n_0$, as in the source proof; then $h < 4 \cdot 2^{-j}$ and $j \rightarrow \infty$ as $h \downarrow 0$. Since $I_\tau \subset [x, x+h]$ and the coordinates are orthogonal,

$$\|F_a(x+h) - F_a(x)\|_2 \geq \left\| (P) \int_{I_\tau} G_a \right\|_2.$$

The n_0 -column alone gives

$$\left\| (P) \int_{I_\tau} G_a \right\|_2 \geq |a_{n_0}| \sqrt{n_0 + 1} 2^{-j/2} \left(\sum_{m=j}^{\infty} \frac{1}{(m+1)^2} \right)^{1/2}.$$

Therefore

$$\left\| \frac{F_a(x+h) - F_a(x)}{h} \right\|_2 \geq \frac{|a_{n_0}| \sqrt{n_0 + 1}}{4} 2^{j/2} \left(\sum_{m=j}^{\infty} \frac{1}{(m+1)^2} \right)^{1/2}.$$

The right-hand side tends to $+\infty$, since the tail sum is at least a constant multiple of $1/(j+1)$. The same argument applies for $h < 0$. Thus the primitive has unbounded norm difference quotients at every point. Weak differentiability at a point would make the corresponding difference quotients weakly convergent, hence norm bounded by uniform boundedness, a contradiction. So $G_a \in \mathcal{ND}_{\ell_2}$.

Now let X contain a subspace Y isomorphic to ℓ_2 , and let $U: \ell_2 \rightarrow Y \subset X$ be an isomorphism. The map

$$a \in \ell_2 \mapsto U \circ G_a$$

is still an isomorphic embedding in the Pettis norm, because

$$\|U^{-1}\|^{-1} \|G_a\|_{\mathcal{P}} \leq \|UG_a\|_{\mathcal{P}} \leq \|U\| \|G_a\|_{\mathcal{P}}.$$

Bounded linear images of Pettis integrable functions are Pettis integrable, and the unbounded norm-difference-quotient estimate survives with the lower constant $\|U^{-1}\|^{-1}$. Hence every nonzero element of this closed infinite-dimensional subspace belongs to $\mathcal{ND}_X$. This proves the theorem.

Verification Notes

The construction uses only the dyadic sets and orthogonal-coordinate calculation already present in the source paper. The new point is the column-wise choice of generators and the $\sqrt{n+1}$ normalization, which turns the Pettis norm estimates into a two-sided ℓ_2 estimate.

The result is partial. It covers Hilbert spaces and all Banach spaces containing an isomorphic copy of ℓ_2 . It does not settle the source problem for arbitrary infinite-dimensional Banach spaces.

Novelty Check

Cheap run indexes were searched for arXiv:1403.1908, title terms, Pettis primitive phrases, and spaceability terms. Web searches for exact title and phrases such as “Pettis primitives spaceable” and “nowhere weakly differentiable Pettis spaceability” found only the source paper. A local full-source search found citations to the source paper and general Pettis-integration references, but no later exact solution of the spaceability problem.

Human Review Recommendation

Review as a likely valid partial result. The main points to check are the Pettis-norm closedness argument and the transfer from ℓ_2 to an isomorphic Hilbert subspace of X .

References

- [1] B. Bongiorno, U. B. Darji, and L. Di Piazza, *Lineability of non-differentiable Pettis primitives*, arXiv:1403.1908.
- [2] J. Diestel and J. J. Uhl, *Vector Measures*, Mathematical Surveys, vol. 15, American Mathematical Society, 1977.